ESD Lab 1 — Setup & Verification Guide

Overview

Install four free tools before the lab: JDK 17, IntelliJ IDEA, MySQL 8.x and Postman. Plan on about an hour, mostly downloads.

#	Tool	Version	Official download	Used for
1	Java JDK	17 exactly	Eclipse Temurin 17 or Oracle JDK 17	Compiling and running the Spring Boot code
2	IntelliJ IDEA	Free version	jetbrains.com/idea/download	Writing and running the project
3	MySQL	8.x	MySQL Community Server (Windows: MySQL Installer)	Database, used from the command line
4	Postman	Latest	postman.com/downloads	Sending test requests to the API

Recommended order: JDK, then IntelliJ, then MySQL, then Postman, then the final check. IntelliJ needs the JDK to exist first.

Before you start

- Use a laptop where you have administrator rights.
- Know your system type. Windows: Settings, System, About, then read System type (64-bit is normal). Mac: Apple menu, About This Mac, then read Chip (Apple M-series or Intel).
- Keep a note of your MySQL root password the moment you set it. You will need it again in the lab.

Step 1 — Java JDK 17

Install JDK 17 (not 11 or 21), set JAVA_HOME, add its bin folder to PATH, then confirm that both `java` and `javac` report version 17.

Download

Pick one distribution. Temurin is free and the simplest choice.

Option	Download page	Notes
Eclipse Temurin 17 (recommended)	adoptium.net/temurin/releases/?version=17	Choose operating system, architecture, package type JDK, version 17
Oracle JDK 17	oracle.com/java/technologies/javase/jdk17-archive-downloads.html	May ask for an Oracle account on some downloads

Which file to pick:

- **Windows:** x64 installer, the `.msi` file.
- **Mac with Apple chip (M1 or newer):** aarch64 `.pkg` file.
- **Mac with Intel chip:** x64 `.pkg` file.

Windows

1. Run the downloaded `.msi` and follow the installer. On the Temurin Custom Setup screen, enable the options to set JAVA_HOME and add Java to PATH if they are offered. You should still check steps 3 and 4 below.
2. Note the install location. Typical paths are `C:\Program Files\Java\jdk-17` (Oracle) or `C:\Program Files\Eclipse Adoptium\jdk-17...` (Temurin).
3. Set JAVA_HOME. Click Start, search for Environment Variables, open Edit the system environment variables, then click Environment Variables. Under System variables click New and enter:
   - Variable name: `JAVA_HOME`
   - Variable value: your JDK 17 install path, the folder that contains `bin`, for example `C:\Program Files\Java\jdk-17`
4. Add Java to PATH. Under System variables select Path, click Edit, click New and add `%JAVA_HOME%\bin`. Click OK on every window.
5. Close any open terminal, then open a new Command Prompt and run:

```
java -version
javac -version
echo %JAVA_HOME%
```

Both version commands must show 17.0.x. The last command must print your JDK path.

macOS

1. Open the downloaded `.pkg` and follow the installer.
2. Open Terminal and list installed JDKs. Version 17 must appear:

```
/usr/libexec/java_home -V
```

3. Set JDK 17 for the current Terminal window:

```
export JAVA_HOME=$(/usr/libexec/java_home -v 17)
```

4. Make it permanent for the default zsh shell. Add the same line to `~/.zshrc`, then reload it:

```
echo 'export JAVA_HOME=$(/usr/libexec/java_home -v 17)' >> ~/.zshrc
source ~/.zshrc
```

5. Verify:

```
java -version
javac -version
echo $JAVA_HOME
```

Both version commands must show 17.0.x. The last command must print a path ending in a 17 folder.

Step 2 — IntelliJ IDEA

Install the free version of IntelliJ IDEA, point its Project SDK at JDK 17, and prove it by running a small Java program.

Download

Go to jetbrains.com/idea/download and download the installer for your system. The free version is enough for this lab, so no licence or paid plan is needed. Mac users must choose the Apple Silicon build for M-series chips or the Intel build for Intel Macs.

Install

1. **Windows:** run the downloaded `.exe`, accept the defaults and finish the wizard.
2. **macOS:** open the `.dmg`, then drag IntelliJ IDEA into the Applications folder.
3. Launch IntelliJ IDEA once to complete its first-run setup.

Point IntelliJ at JDK 17

1. Create or open any project.
2. Open File, then Project Structure, then Project.
3. Under Project SDK, choose JDK 17.
4. If JDK 17 is not in the list, open the dropdown, choose Add SDK, then Add JDK, and select the JDK 17 folder you installed in Step 1 (the same folder as JAVA_HOME).
5. Click Apply, then OK.

Verify

1. Create a new Java project (File, New, Project) and select JDK 17 as the SDK.
2. Add a class with a `main` method that prints one line.
3. Click the green Run arrow.

It is working when the project shows JDK 17 as its SDK and the program builds and prints its line in the Run window.

Step 3 — MySQL 8.x

Install MySQL 8.x with a root password you will remember, then log in from the command line and read back the version. MySQL Workbench is not needed.

Download

System	Download page	What to pick
Windows	MySQL Installer	The larger installer file (offline, `mysql-installer-community`), which does not need internet while installing
macOS	MySQL Community Server	The `.dmg` for your chip: ARM, 64-bit for Apple M-series, or x86, 64-bit for Intel

On the download page, open the Select Version dropdown and choose an 8.x release, for example 8.4 LTS. Avoid the 9.x versions. If the page asks you to log in, click No thanks, just start my download.

Windows

1. Run the MySQL Installer.
2. Choose Server only. A Custom setup works too, as long as MySQL Server is selected.
3. Click Next and Execute to install, then continue through the configuration screens with the defaults.
4. When asked for Accounts and Roles, set a password for the root user. Write it down. You will use it in the Spring Boot configuration.
5. Finish the configuration. Leave MySQL running as a Windows service when offered.
6. Open a new Command Prompt and check the client:

```
mysql --version
```

7. Connect, entering your root password when prompted:

```
mysql -u root -p
```

8. Inside MySQL, test the server, then leave:

```
SELECT VERSION();
exit
```

The version must start with 8. If `mysql` is not recognized, add the MySQL bin folder to PATH. A typical location is `C:\Program Files\MySQL\MySQL Server 8.0\bin` (the folder name matches your version, for example `MySQL Server 8.4`). Add it under System

variables, Path, Edit, New, then open a new Command Prompt.

macOS

1. Open the downloaded `.dmg` and run the `.pkg` inside it.
2. When the installer asks for authentication, choose the recommended strong password encryption, then set a root password. Write it down.
3. Open Terminal and check the client:

```
mysql --version
```

4. Connect, entering your root password when prompted:

```
mysql -u root -p
```

5. Inside MySQL, test the server, then leave:

```
SELECT VERSION();
exit
```

The version must start with 8. If the `mysql` command is not found, add the MySQL bin folder to PATH. The usual location is `/usr/local/mysql/bin`. Run:

```
echo 'export PATH="/usr/local/mysql/bin:$PATH"' >> ~/.zshrc
source ~/.zshrc
```

If MySQL will not connect

- Access denied: the password is wrong. Retype it carefully, as nothing shows on screen while you type.
- Can't connect to server: MySQL is not running. On Windows, open Services and start the MySQL service. On Mac, open System Settings, then MySQL, and click Start MySQL Server.

Step 4 — Postman

Install Postman, open it once, and confirm you can create a new HTTP request. The Lab 1 repository will supply the Postman collection used in the session.

Download

Go to postman.com/downloads and download the installer for your system. For Mac, pick Apple Silicon for M-series chips or Intel for Intel Macs.

Install

1. **Windows:** run the downloaded `.exe`. Postman installs itself and opens when done.
2. **macOS:** open the downloaded file, then drag Postman into the Applications folder and launch it.
3. On first launch, Postman may ask you to sign in or create a free account. You can skip it with the Skip and go to the app option, or sign up if you prefer.

Verify

1. Open Postman.
2. Click New, or the plus sign for a new tab, and choose HTTP.
3. Confirm that a blank request screen opens with a URL box and a method dropdown (GET, POST and others).

It is working when Postman opens and you can create a new HTTP request.

Step 5 — Final verification

Run these commands in a fresh terminal. Java must report 17 and MySQL must report 8.x.

Windows (Command Prompt)

```
java -version
javac -version
mysql --version
```

Then test the MySQL login:

```
mysql -u root -p
SELECT VERSION();
exit
```

macOS (Terminal)

```
java -version
javac -version
mysql --version
```

Then test the MySQL login:

```
mysql -u root -p
SELECT VERSION();
exit
```

What a pass looks like

Command	Expected result
`java -version`	Shows version 17.0.x
`javac -version`	Shows `javac 17.0.x`
`mysql --version`	Shows a version starting with 8
`mysql -u root -p`	Asks for the password, then shows the `mysql>` prompt
`SELECT VERSION();`	Returns a value starting with 8

Step 6 — Ready for the lab?

Tick every item. If all are checked, your system is ready for ESD Lab 1.

- [ ] JDK 17 installed
- [ ] JAVA_HOME configured
- [ ] `java -version` shows Java 17
- [ ] `javac -version` shows Java 17
- [ ] IntelliJ IDEA installed and can use JDK 17
- [ ] MySQL 8.x installed
- [ ] `mysql --version` works
- [ ] `mysql -u root -p` works

- [ ] MySQL root password is known
- [ ] Postman installed and opens successfully

Common problems and fixes

Problem	Fix
`java -version` shows a version other than 17	Another JDK is first on PATH. On Windows, move `%JAVA_HOME%\bin` to the top of the Path list. On Mac, rerun the `export JAVA_HOME` line and check `~/.zshrc`.
`java` or `javac` is not recognized	You did not open a new terminal after setting variables, or `%JAVA_HOME%\bin` is missing from Path. Close the terminal, reopen it and recheck.
JAVA_HOME points to the `bin` folder	It must point to the JDK folder one level above `bin`.
JDK 17 missing in IntelliJ's SDK list	Use Add SDK, then Add JDK, and select the JDK 17 install folder.
`mysql` is not recognized or not found	Add the MySQL bin folder to PATH (see Step 3), then open a new terminal.
Access denied for user root	The root password is wrong. Retype it carefully.
Can't connect to MySQL server	The MySQL service is not running. Start it from Windows Services or the Mac System Settings.